

Body barriers

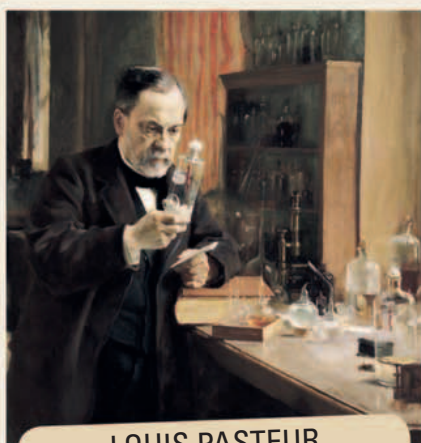
Germes such as bacteria and viruses are always trying to get into your body. Fortunately, the human body is well protected. The first line of defense against invaders is the body's surface, which acts as a barrier to germs. The surface includes not only the tough, outer layer of skin but also the delicate surface of your eyes and the soft tissues lining your mouth, nose, throat, and stomach.

INNER BARRIERS

Germes can enter your body through the air you breathe in or through the food you eat. To stop these germs from getting any farther, the cells lining inner passages such as the throat and airways are packed together to form a germproof wall. This wall is covered in protective fluids such as saliva, mucus, and stomach acid.

► **STICKY STUFF** *A sticky fluid called mucus, seen here magnified (green) covers the inner walls of airways leading to the lungs. Germs that land on mucus get stuck. Tiny hairs (pink) beat back and forth to push the mucus up to the throat, where it gets swallowed.*

The salivary glands make saliva, which contains chemicals that kill bacteria.



LOUIS PASTEUR

French scientist Louis Pasteur (1822–95) was the first person to realize that disease can be caused by tiny organisms—germs. Before Pasteur, most people thought diseases were a punishment from God.



▲ **STOMACH GLAND** *The stomach's wall contains glands that produce a very powerful acid. This acid destroys germs in food. It also kills the germs in mucus from the throat, which we swallow.*